

The Role of AI Literacy in Trust Calibration toward AI-Generated Content

Revised Questionnaire

Section 1. Demographics and background

1.1. What is your age? ____

Source: Brown, Sillence & Branley-Bell (2025), *AcademAI: Investigating AI usage, attitudes, and literacy in higher education and research*. *Journal of Educational Technology Systems*, 54(1), 6–33.

1.2. What is your gender?

Female / Male / Non-binary or other / Prefer not to say

Source: Brown, Sillence & Branley-Bell (2025), *AcademAI*.

1.3. What is your main field of study?

Computer Science/Informatics; AI/Data Science/Machine Learning; Engineering/Technology; Mathematics/Statistics; Natural Sciences; Social Sciences; Economics/Econometrics; Business/Management/Marketing; Health/Medicine/Psychology; Communication/Media/Journalism; Arts/Design; Other: ____

1.4. What is your current academic status?

Undergraduate / Master's student / PhD student / Graduate working / Other

Source: Brown, Sillence & Branley-Bell (2025), *AcademAI*.

1.5. Have you taken any courses related to AI, machine learning, or data science?

Yes (>3 courses) / Yes (2–3 courses) / Yes (1 course) / No

Source: KPMG International & University of Melbourne (2025). *Trust, attitudes and use of artificial intelligence: A global study 2025*.

1.6. How long have you been actively using AI tools such as ChatGPT, Copilot, or Gemini?

Less than 6 months / 6–12 months / 1–2 years / More than 2 years / I don't use them

Section 2. AI usage behavior

2.1. How often do you use AI tools such as ChatGPT, Copilot, or Gemini?

Multiple times per day / Daily / A few times a week / A few times a month / Less than once a month / Never or almost never

Source: Brown, Sillence & Branley-Bell (2025), *AcademAI*. *Journal of Educational Technology Systems*, 54(1), 6–33.

2.2. In a typical week, approximately how much time do you spend using AI tools?

More than 7 hours / 3–7 hours / 1–3 hours / 30–60 minutes / Less than 30 minutes / I don't use them weekly

2.3. Overall, how confident do you feel in your ability to use AI tools effectively? (1 = Not confident at all, 5 = Extremely confident)

Source: Department for Science, Innovation and Technology & Department for Culture, Media and Sport (2026). AI skills for life and work: General public survey findings. GOV.UK. / KPMG International & University of Melbourne (2025). Trust, attitudes and use of artificial intelligence: A global study 2025.

2.4. ATTENTION CHECK. What is 4 + 4? Please type the number.

Source: Standard attention check, Oppenheimer, Meyvis & Davidenko (2009), JESP.

2.5 To what extent do you feel you know about AI? (1=my knowledge about AI is low, 5= my knowledge about AI is high)

Source: KPMG International & University of Melbourne (2025). Trust, attitudes and use of artificial intelligence: A global study 2025.

Section 3. Objective AI knowledge

These questions are about how AI systems work. There is one correct answer per question. Please answer based on what you actually know — selecting 'I don't know' is always fine.

Source: All items reproduced from Hornberger, Bewersdorff & Nerdel (2023), Computers and Education: AI, Appendix.

Machine learning fundamentals

3.1. What is a key criterion for the quality of a model in machine learning?

a) It can predict the output values of the test data as well as possible [correct] / b) It contains as few variables as possible / c) It is as well adapted as possible to the training data / d) The predictions are as unambiguous as possible / e) I don't know

3.2. How does supervised learning differ from unsupervised learning?

a) In supervised learning, the output values of the training data are known [correct] / b) In supervised learning, humans must supervise the AI during learning and intervene if necessary / c) In supervised learning, all computational steps are documented / d) In supervised learning, stricter legal regulations apply / e) I don't know

3.3. What should be considered in machine learning when dividing the data into training and test data?

a) The data should be divided into parts of as equal size as possible / b) The data should be randomly divided into training and test data sets [correct] / c) The test data should be of higher quality than the training data / d) The training and test data should be as different from each other as possible / e) I don't know

Human role in AI

3.4. How can humans influence the outcome of machine learning?

a) Calculation of the accuracy of the prediction / b) Randomized division into test and training data / c) Selection of the model [correct] / d) Abstraction of the model / e) I don't know

3.5. To what extent can humans influence the outcome of machine learning?

a) Humans can hardly influence the result of machine learning, since it runs automatically / b) The results can only be influenced when selecting the data the model learns with / c) Humans can influence the result during development at several process steps [correct] / d) Humans can only influence the interpretation of the results / e) I don't know

How AI works

3.6. What determines the behavior of AI systems?

a) AI systems strive for autonomy / b) AI systems pursue a goal that has been given to them by humans [correct] / c) AI systems perform behaviors randomly / d) AI systems seek out goals independently and pursue them / e) I don't know

3.7. Why can systems based on machine learning obtain good results?

a) Their work is often observed by humans and corrected if necessary ('supervised learning') / b) They think similarly to humans, but faster / c) They can draw conclusions from large amounts of data and thus improve their model [correct] / d) They are derived from expert systems in which expert knowledge is stored / e) I don't know

3.8. What data do AI-based recommender systems of streaming services use?

a) The data of every user when using the service [correct] / b) All data that a user of the service leaves on the internet / c) The data of other users, but not one's own / d) Only one's own data when using the service / e) I don't know

Limitations and ethics

3.9. You are testing a machine learning model that is supposed to classify photos of animals. You notice that the model is better at recognizing cats than dogs. What could be the reason for this?

a) Dogs are more difficult to recognize than cats, since there are fewer images of dogs on the internet / b) Small objects, such as cats, are better recognized than large ones, such as dogs / c) Most models are generally better at recognizing cats than dogs / d) The training data for dogs were not representative of all dog breeds [correct] / e) I don't know

3.10. What is the black box problem?

a) AI entails a residual risk that is hard to calculate / b) It is often difficult to determine how AI makes decisions [correct] / c) Users are often not informed about the application of AI / d) Many users have little knowledge about AI / e) I don't know

3.11. What are central risks in using AI for predictive policing?

a) Vulnerability to hacking / b) Discrimination against suspects based on origin and status [correct] / c) Lack of legal certainty in the event of AI breakdown / d) Undermining the authority of police officers / e) I don't know

Self-estimate of performance (calibration measure)

3.12. Of the previous 11 questions, approximately how many do you think you answered correctly? (0–11) ____

Source: Calibration item following Welsch et al. (2025), Computers in Human Behavior. Overconfidence score = self-estimate minus actual OLS. Positive = overconfident. Reported descriptively.

Section 4. Trust in AI tools

Please indicate your agreement with the following statements. (1 = Strongly disagree, 5 = Strongly agree)

Competence Trust

- 4.1. The AI tools are reliable.
- 4.2. I think that AI tools are competent and perform their tasks well.
- 4.3. I think that AI tools perform their role very well.
- 4.4. I believe AI tools can do everything I would expect them to.
- 4.5. I can trust the information presented to me by AI tools.
- 4.6. AI tools are fast and efficient.

Perceived Risk

- 4.7. The AI tools are deceptive. (R)
- 4.8. The AI tools behave in an underhanded manner. (R)
- 4.9. I am suspicious of the AI tools' intent, action, or outputs. (R)
- 4.10. The AI tools' actions could lead to harmful outcomes. (R)
- 4.11. It is risky to interact with AI tools. (R)
- 4.12. ATTENTION CHECK. Please select 'Agree' for this item.

Section 5. Domain-specific trust

How much would you trust AI in each of the following situations? (1 = Not at all, 5 = Completely)

Source: Nguyen, N. B., Trinh, Q. M., Luong, H. T., & Nguyen, M. H. (2025). Learning with algorithms: AI usage, trust, and literacy among university students. *Ho Chi Minh City Open University Journal of Science – Social Sciences*, 16(10).

- 5.1. General knowledge questions (e.g., factual lookups)
- 5.2. Learning or study-related tasks
- 5.3. Writing academic work (essays, papers, reports)
- 5.4. Programming or technical tasks
- 5.5. Professional or work-related tasks
- 5.6. Health-related questions
- 5.7. Personal advice or life decisions
- 5.8. News and current events

Appendix A. Construct map

Items	Construct	Source	Used for
1.1–1.2	Age and gender	Brown, Sillence & Branley-Bell (2025), AcademAI, Journal of Educational Technology Systems	Controls
1.3–1.4	Field of study and academic status	Developed for this study; consistent with Hornberger et al. (2023)	Controls
1.5	AI course experience	KPMG International & University of Melbourne (2025), Global AI Trust Study	Control variable
1.6	AI usage duration	Developed for this study	Control variable
2.1	AI use frequency	Brown et al. (2025), AcademAI	Control variable
2.2	Weekly AI time	Developed for this study	Control variable
2.3	Self-rated AI confidence	DSIT (2026), AI skills for life and work; KPMG (2025)	Exploratory — overconfidence gap
3.1–3.11	Objective AI literacy	Hornberger, Bewersdorff & Nerdel (2023), CEEdAI, 3-PL IRT validated. Items reproduced from Appendix, Table 1/2.	Main IV — H1 and H2
3.12	Self-estimated performance	Welsch et al. (2025), Computers in Human Behavior	Exploratory — overconfidence gap
4.1–4.6	Competence Trust	Sun et al. (2025) HAITs, Table 3. Verbatim items. Omega = .966, AVE = .626	H1, H2 (primary DV)
4.7–4.11	Perceived Risk (R)	Sun et al. (2025) HAITs, Table 3. Verbatim items. Omega = .959, AVE = .683	H1, H2 (primary DV)
5.1–5.8	Domain-specific trust	Developed for this study; informed by Nguyen et al. (2025) and KPMG (2025)	Descriptive — domain calibration pattern

References

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